

MTH205 Analysis in one variable

Tutorial 2

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- (1) Find an open cover of $\mathbb{N}$ that has no finite sub-cover.

Solution: For each $n \in \mathbb{N}$, take $G_n := (n - \frac{1}{2}, n + \frac{1}{2})$ then $\mathcal{G} = \{G_n : n \in \mathbb{N}\}$ is collection of open subsets of $\mathbb{R}$ and it is an open cover for $\mathbb{N}$. Clearly, there is no finite sub collection of $\mathcal{G}$ that cover $\mathbb{N}$.

- (2) If $\{K_n : n \in \mathbb{N}\}$ is a sequence of non-empty compact sets in $\mathbb{R}$ such that

$$K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots K_n \supseteq \cdots,$$

then prove that $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$.

Solution: For each $n \in \mathbb{N}$, choose $x_n \in K_n$. Here, the sequence $\{x_n\}_{n=1}^{\infty}$ in the compact set K_1 and hence $\{x_n\}_{n=1}^{\infty}$ is bounded. By the Bolzano-Weierstrass theorem, there is a subsequence $\{x_{n_r}\}$ of $\{x_n\}$ such that $\{x_{n_r}\}$ converges to a point x_0 . Since $x_{n_r} \in K_n$ for all $r \geq n$. This follows that $x_0 \in \bigcap_{n=1}^{\infty} K_n$.

- (3) Let K be a non-empty compact subset of $\mathbb{R}$. Then show that both $\inf K$ and $\sup K$ exist and belong to K .

Solution: Since K is a non-empty compact subset of $\mathbb{R}$. Clearly, K is bounded and so, $\sup K$ exists in $\mathbb{R}$. For each $n \in \mathbb{N}$ there is an $y_n \in K$ with $y_n + \frac{1}{n} \geq \sup K$. We can see that $K_n := \{x \in K : x + \frac{1}{n} \geq \sup K\}$ is a non-empty closed, bounded and hence compact.

Moreover, $K_1 \supset K_2 \supset K_3 \supset \cdots$, and as a result $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$. If $x_0 \in \bigcap_{n=1}^{\infty} K_n$ then $x_0 \in K$ and

$$x_0 + \frac{1}{n} \geq \sup K \geq x_0, \text{ for all } n \in \mathbb{N}.$$

Therefore, $x_0 = \sup K \in K$. Similarly, one can show that $\inf K$ exist and belong to K .

- (4) Give an example of a disjoint closed subsets F_1, F_2 of $\mathbb{R}$ such that

$$0 = \inf \{|x_1 - x_2| : x_1 \in F_1, x_2 \in F_2\}.$$

Solution: Let $F_1 = \mathbb{N}$, $F_2 = \{n + \frac{1}{n} : n \geq 2\}$. Then F_1, F_2 are closed disjoint sets and

$$\inf \{|x_1 - x_2| : x_1 \in F_1, x_2 \in F_2\} = 0.$$

- (5) Is Cantor set a nowhere dense set ? Justify your answer.

Solution: The Cantor set is nowhere dense as it contains no open interval in it.

- (6) Prove the following:

- (a) If $E \subset \mathbb{R}$ then E is nowhere dense if and only if $\mathbb{R} \setminus \overline{E}$ is dense subset of $\mathbb{R}$.
- (b) $\mathbb{R}$ can not be written as a union of countable collection of nowhere dense sets.

Solution: (a) Suppose that $\mathbb{R} \setminus \overline{E}$ is dense in $\mathbb{R}$ then $\overline{\mathbb{R} \setminus \overline{E}} = \mathbb{R}$. Note that $x \in \overline{\mathbb{R} \setminus \overline{E}}$ if and only if for every $\epsilon > 0$ we have $(x - \epsilon, x + \epsilon) \cap \mathbb{R} \setminus \overline{E} \neq \emptyset$ if and only if $x \in \mathbb{R} \setminus \text{int}(\overline{E})$. As a consequence, $\mathbb{R} \setminus \text{int}(\overline{E}) = \mathbb{R}$. Thus $\text{int}(\overline{E}) = \emptyset$. The reverse argument will establish the forward implication.

Solution: (b) The result follows from the “Baire category theorem”.

Definition 0.1. Let $A \subseteq \mathbb{R}$. Then a point $x \in A$ is said to be *interior point* if there exists $\epsilon > 0$ such that

$$\mathcal{N}_\epsilon(x) = (x - \epsilon, x + \epsilon) \subseteq A.$$

A point $y \in \mathbb{R}$ is called a *cluster point* or *limit point* of A if for every $\epsilon > 0$, the set $\mathcal{N}_\epsilon(y) \setminus \{y\} \cap A$ is non-empty.

The set of all interior points of A is denoted by $\text{int}(A)$ and the collection of all limit points of A is denoted by $\overline{A}$.

Note that $\text{int}(A)$ is the largest open set contained in A . Whereas, $\overline{A}$ is the smallest closed set containing A .

Definition 0.2. A subset A of $\mathbb{R}$ is said to be

- *dense set* if $\overline{A} = \mathbb{R}$.
- *nowhere dense* if $\text{int}(\overline{A}) = \emptyset$.